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1 SEM MCA (R) (CBCS) MATHS

2021

COMPUTER APPLICATION

Paper: MCA 108

(*Discrete Mathematics*)

Full Marks: 30

Time: 90 minutes

Mode of Examination: Online (Open Book Examination)

Answer any six from the following

1. If X is a non empty set such that X={1,2,3,4} and 5

$\bar{A} = \left\{\frac{1}{.2}, \frac{2}{.4}, \frac{3}{.6}, \frac{4}{.9}\right\}$ and $\bar{B} = \left\{\frac{1}{.3}, \frac{2}{.5}, \frac{3}{.7}, \frac{4}{1}\right\}$ are two fuzzy sets on X then find the following

(i) $\bar{A} \cup \bar{B}$ (ii) $\bar{A} \cap \bar{B}$ (iii) height of $\bar{A}$ and $\bar{B}$ (iv) ${}^{\alpha}\bar{A}$ when α=.4 (v) Support of $\bar{A}$ and $\bar{B}$

2. If $U = \{1,2,3,4,5,6\}$, $A = \{1,2,3,4\}$ and $B = \{3,4,5,6\}$. Find the bit string for the set A and B and use bit string to find A' and union and intersection of A and B. 5

3. Evaluate $\lceil \lfloor -4.3 \rfloor - \lfloor -3.6 \rfloor \rceil - \lfloor \lceil -5.2 \rceil - \lceil -1.5 \rceil \rfloor$ 5

4. Let P(S) be the power set of a given set S. Show that the inclusion relation ⊆ is a partial ordering on the power set P(S). 5

5. Find the number of permutations that can be formed by the letters of the word 'DAUGHTER' (i) taking all the letters together without restriction (ii) beginning with D (iii) beginning with D and ending with R (iv) vowels being always together (v) vowels occupying even places. 5

6. Let A be the set of non- zero integers and let R be the relation on $A \times A$ defined by

$$(a, b)R(c, d) \Leftrightarrow ad = bc$$

Show that R is an equivalence relation. 5

7. Obtain the CNF of the following 5

$$[q \vee (p \wedge r)] \wedge \sim[(p \vee r) \wedge q]$$

8. Consider the subsets {2,3}, {4,6} and {3,6 } in the poset ({1,2,3,4,5,6}, /),find for each subsets, if exists

(i) upper bound and lower bound (ii) greatest lower bound and least upper bound 5